

SOURAV KUMAR

+91 7463959457 | 2204rajsourav@gmail.com | <https://www.linkedin.com/in/sourav-kumar-736a8a286/>
<https://github.com/souravkumar1414>

EDUCATION

University of Petroleum and Energy Studies (UPES), Dehradun
B. Tech, Computer Science & Engineering

May 2026
CGPA: 7.36/10

SKILLS

Programming Languages: Python, SQL

Machine Learning & Deep Learning: Scikit-learn, Supervised Learning (Regression, Classification), Linear Regression, Decision Tree, Random Forest

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA)

Model Evaluation: MAE, MSE, RMSE, R² Score, Feature Engineering, Hyperparameter Tuning

Tools: Jupyter Notebook, Google Colab, Git, GitHub, VS Code

EXPERIENCE

Intern | IBM | Remote

June 2025 - July 2025

- Assisted in collecting and preprocessing structured healthcare datasets. Supported exploratory data analysis (EDA) to uncover cost-driving factors.
- Aided in implementing regression models including linear regression, decision tree, and random forest.
- Built a predictive system to estimate medical insurance charges.
- Visualized insights using Matplotlib/Seaborn.

Social Work Intern | Samadhaan Foundation

June 2024 - Aug 2024

- Assisted in community outreach programs.
- Conducted data analysis and helped in digital management initiatives.

PROJECTS

Stock Prediction Models

Aug 2025.

- Developed multiple stock price prediction models using Machine Learning and Deep Learning techniques, improving forecasting accuracy by evaluating and comparing model performance across algorithms such as LSTM, ARIMA, and Prophet.
- Performed data preprocessing, feature engineering, and normalization to improve prediction accuracy.
- Implemented visualization dashboards using Matplotlib and Seaborn for exploratory data analysis (EDA), enabling insights into historical trends, volatility, and model performance metrics.
- Conducted hyperparameter tuning and cross-validation to optimize models, resulting in more stable forecasting outcomes and reduced overfitting across test scenarios.

CineScope - Movie Discovery App

Dec 2024

- Designed a responsive, single-page app using React 19, Tailwind CSS, and Vite, enabling users to browse movie details via TMDB API, with a clean UI optimized for engagement.
- Defined a custom “Trending” feature by integrating Appwrite as a lightweight backend, demonstrating product thinking.
- Optimized the search solution by debouncing the input field, reducing redundant API calls by over 70% during active input and improving performance, secure queries through *encodeURIComponent*; manually hosted via AWS Amplify.

NeuroFin - Classification Tool (Transfer Learning)

May 2024

- Trained an image classification model using transfer learning with MobileNetV3-Large and a dataset of 9,000 images to accurately classify 9 fish species with 90% overall accuracy and reducing reliance on manual inspection.
- Enhanced model robustness and usability through Python-based preprocessing, augmentation pipelines, and splitting (80/20 train-validation) ensuring scalable deployment across varied datasets.

CERTIFICATIONS

- Data Structures and Algorithms (DSA) Certification – Coursera
- IBM Certified Internship – Healthcare Cost Prediction
- Python Programming – CodeWithHarry (Self Learning)
- Harvard CS50AI: CS50’s Introduction to Artificial Intelligence with Python
- Microsoft Certified: Azure AI Fundamentals (AI-900)

KEY ACHIEVEMENTS

- Member of the Coding Club at UPES
- Volunteered at TechFest 2023, UPES
- Active participant in hackathons and coding challenges